

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (CSE) (Credit System) Degree Examinations
May - June 2023

20CS601/ 17CS704 - MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Define Machine Learning. Write a note on Machine Learning versus Traditional programming.	6	L*2	1	1
	b) What is a learning program? Explain the symbols E, T and P in a learning system in application to a machine learning problem.	6	L2	1	1
	c) Explain the components of learning process in detail.	8	L3	1	2
2.	a) Explain the seven steps of Machine Learning in detail.	10	L2	1	1
	b) Discuss about the classification of Machine Learning algorithms. List any 6 applications of Machine Learning.	10	L3	1	1
Unit – II					
3.	a) What is Decision tree learning? Discuss the appropriate problems for Decision tree learning.	6	L2	2	1
	b) Explain ID3 algorithm in detail.	6	L2	2	2
	c) Explain Back propagation algorithm in detail.	8	L2	2	2
4.	a) Write the appropriate problems for neural network learning.	6	L2	2	1
	b) What is perceptron? Explain the perceptron training rule.	6	L2	2	2
	c) Explain the working of SVM. List any 4 advantages and disadvantages of SVM.	8	L2	2	2
Unit – III					
5.	a) Illustrate Naive Bayes Classifier with an example.	10	L3	3	2
	b) Discuss KNN algorithm. List the advantages and disadvantages of KNN.	10	L2	3	1
6.	a) Compare linear and logistic regression.	10	L2	3	2
	b) Briefly explain Random forest algorithm. List the advantages, disadvantages and applications of random forest algorithm.	10	L1	3	1
Unit – IV					
7.	a) Explain K-means clustering algorithm in detail.	10	L2	4	2
	b) Discuss the various Model evaluation and classification Metrics (any 5).	10	L2	4	2
8.	a) Discuss Apriori Algorithm.	6	L1	4	1
	b) Explain Hierarchical clustering in detail.	8	L2	4	1
	c) Differentiate Supervised and Unsupervised Machine Learning algorithms.	6	L2	4	1
Unit – V					
9.	a) Explain Markov Decision Process (MDP) in detail with an example.	10	L2	5	1
	b) Illustrate Q Learning algorithm with an example.	10	L2	5	2
10.	a) Explain Reinforcement learning.	7	L1	5	1
	b) Write a note on semi supervised learning.	5	L1	5	1
	c) Discuss Non deterministic Rewards and actions.	8	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
